# Reviewer-Proofing the Design
# Toward a Top-Journal Contribution

*A critical upgrade of node design, edge design, and the forecasting pipeline*

Graph-based forecasting of Google Trends agricultural search interest in Florida

Companion to the literature review · Target: International Journal of Forecasting / ISF

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# Contents

## 0. The Honest Diagnosis

**Your literature review is genuinely good** — the matrix, the gap statement, and the Design-A starting point are all sound, and the reference set (MTGNN, AGCRN, Graph WaveNet, StemGNN, SDGL, plus the Trends-calibration papers) is exactly right. The problem is not what is there; it is that the design as written is a *competent applied project*, and competent applied GNN-on-a-new-dataset papers are routinely rejected at highly regarded forecasting venues. They get rejected for a predictable set of reasons, and every one of them is fixable in advance.

> **The reviewer's two killer questions.** (1) *"You forecast a noisy, relative, resampled attention index slightly better than ARIMA — so what? Why does anyone care about the number?"* (2) *"Your GNN beats your LSTM, but the GNN has more capacity and sees all series at once. How do you know the improvement comes from the graph structure and not just from extra parameters and cross-series pooling?"* A top-journal version of this project is essentially the design that answers these two questions before they are asked.

This document does three things: (1) names each gap between the current design and the top-journal bar; (2) gives an upgraded node and edge design that is more novel *and* more defensible; and (3) specifies a pipeline whose results survive adversarial review. Section 8 reframes the contribution itself, which is the single highest-leverage change.

## 1. Gaps Between the Current Design and the Top-Journal Bar

| Gap | Why it sinks the paper | Fix (detailed later) |
|---|---|---|
| No capacity-matched control | GNN beating LSTM/VAR may reflect more parameters + cross-series pooling, not graph structure. This is the #1 overclaim in GNN-forecasting papers. | Add a GLOBAL no-graph model with matched capacity, AND a graph-permutation null (§4). |
| No graph-permutation / random-graph null | Without it you cannot attribute gains to topology. Reviewers now demand it. | Re-run the SAME architecture with shuffled / random / identity adjacency (§4). |
| No significance testing | Raw MAE/RMSE tables are not evidence at IJF. Differences may be noise. | Diebold–Mariano per pair; Friedman + MCB/Nemenyi across series (§5). |
| Target-validity ("so what") not addressed | Forecasting a noisy attention index has no stakes on its own. | External validity: link Trends to USDA shipments/prices; nowcasting framing (§8). |
| Trends sampling noise ignored as a target property | The dependent variable is itself random; you may be 'beating' models below the data's own noise floor. | Multi-sample pulls; estimate the irreducible noise floor; report skill relative to it (§3, §6). |
| Correlation graph on raw series | Two seasonal series correlate because both follow the calendar — the graph just re-encodes seasonality your Fourier features already supply. | Deseasonalize first; prefer PARTIAL correlation / graphical lasso (§3). |

| Gap | Why it sinks the paper | Fix (detailed later) |
|---|---|---|
| MAPE/sMAPE on 0–100 bounded data | Unstable/undefined near low-interest weeks; not comparable across crops. | Use MASE / RMSSE (scaled vs seasonal-naïve) as primary (§5). |
| 12-week input window vs annual seasonality | A 12-week window cannot see the yearly cycle that dominates these series. | Long input window and/or explicit seasonal features; seasonal-naïve as the bar (§6). |
| Rolling stats / scalers leakage risk | Rolling mean/std and per-series scalers fit across the whole sample leak the future. | Causal, train-fit-only transforms; refit per rolling-origin fold (§6). |
| Only ~5 seasonal cycles | Weekly data over 5 yr ≈ 5 annual cycles — deep models overfit; seasonal-naïve is brutally strong. | Regularization, global training, probabilistic output, honest small-data framing (§6). |

## 2. Upgraded Node Design

Keep **one crop = one node** as the backbone — it is interpretable and correct. But "one crop, one Trends series" is thin, and thinness reads as low novelty. Three upgrades add depth and a genuine contribution without breaking feasibility.

### 2.1 Upgrade A — Search-intent decomposition (recommended novelty)

A single crop term mixes economically distinct demand signals. "strawberry" aggregates **consumption intent** ("strawberry recipe", "strawberry shortcake"), **acquisition/local intent** ("strawberry picking near me", "u-pick"), **price intent** ("strawberry price"), and **production intent** ("growing strawberries", "strawberry plants"). These lead and lag differently and map onto different sides of the market.

> **Why this is publishable:** decomposing each crop node into intent-channels (via Trends *related queries* or curated query groups) turns a flat crop graph into a **crop × intent** graph in which you can ask whether *production-side* attention leads *consumption-side* attention, or whether price-intent is the true leading indicator. That is a demand-economics question, not just a forecasting-accuracy question — exactly the kind of substance reviewers reward.

### 2.2 Upgrade B — Multi-view node features (not multi-node)

Keep the node set = crops, but give each node a multi-channel feature tensor so the model sees more than its own past Trends value:

- **Trends target + causal lags** (1, 2, 4, 8, 13, 26, 52 wks — include the annual lag).
- **Fourier calendar terms** (annual + sub-annual harmonics) and crop-calendar in/out-of-season flags (UF/IFAS).
- **Independent corroborating signal**: Wikipedia pageviews for the crop — a free, fully reproducible second view that helps separate true interest from Trends sampling noise.
- **Exogenous**: Florida weekly weather (NOAA/PRISM); optionally USDA price/shipment lags as features.

- **Static node attributes**: crop family, season class, perishability — used by attention/relational layers and for community validation.

### 2.3 Upgrade C — Hierarchical / heterogeneous extension (second paper)

Promote the heterogeneous graph (your Design C) to a **second, clearly-scoped contribution** rather than an afterthought: a hierarchical graph with crop nodes nested under season/region/commodity-group nodes (HiSTGNN-style). Do NOT try to do everything in one paper — a focused crop(-intent) graph done rigorously beats a sprawling heterogeneous graph done loosely.

## 3. Upgraded Edge Design

Your edge menu (correlation, lagged, Granger, DTW, season/category, learned, hybrid) is complete but *statistically naïve* in two ways reviewers will catch. Fixing both is what separates a credible graph from a decorative one.

### 3.1 Problem 1 — Spurious edges from shared seasonality

> **Almost every pair of Florida crop series is positively correlated** simply because they all rise in the growing/eating season. A raw-correlation graph is therefore mostly a *seasonality graph* — and your Fourier features already encode seasonality, so the graph adds little. **Fix: build edges on deseasonalized, stationarized residuals** (STL remainder, or seasonal differencing), and prefer **partial correlation / graphical-lasso (precision-matrix) edges**, which condition out common drivers and keep only direct dependence. This is the principled version of your reference [10] and should be your statistical backbone, not a side baseline.

### 3.2 Problem 2 — Edges reported without uncertainty

Any correlation/Granger value on ~250 weekly points has wide error bars. Reporting a thresholded graph as if it were ground truth invites the "your graph is noise" critique. **Fix: statistically validated edges.** Keep an edge only if it survives a null model — bootstrap/Politis–Romano stationary-bootstrap confidence intervals, or surrogate-series tests (phase-randomized / IAAFT surrogates) — with multiple-testing control (FDR). This directly echoes the "statistically validated knowledge graph" idea (your [11]) and produces a graph you can defend edge-by-edge.

### 3.3 The upgraded edge menu

| Edge type | Definition / method | Role |
|---|---|---|
| Partial correlation (graphical lasso) | Sparse precision matrix on deseasonalized residuals; edge = direct conditional dependence. | PRIMARY statistical graph — removes spurious shared-season edges. |
| Deseasonalized Pearson/Spearman | Correlation of STL remainders; threshold + top-k. | Interpretable contrast vs partial correlation. |
| Lagged cross-correlation (directed) | argmax_k corr(x_i(t-k), x_j(t)); store lag → directed edge. | Lead–lag structure; the strawberry→blueberry question. |

| Edge type | Definition / method | Role |
|---|---|---|
| Transfer entropy / nonlinear directed info | Model-free directed information; nonlinear generalization of Granger. | Robust directed edges where Granger is unstable on noisy weekly data. |
| Granger causality (VAR-based) | F-test of predictive improvement; on stationarized series. | Econometric directed edge; report with TE for triangulation. |
| DTW shape similarity | 1/(1+DTW) on z-normalized series, Sakoe–Chiba band; top-k. | Phase-shifted seasonal-shape similarity. |
| Agronomic / domain (UF-IFAS + USDA) | Expert edges: shared season window, commodity group, region, substitute/complement. | Knowledge prior; the interpretability anchor. |
| Semantic / co-search | Trends related-query overlap or name-embedding cosine. | Behavioral relatedness beyond co-movement. |
| Learned / adaptive (MTGNN/AGCRN/GWN) | End-to-end node-embedding adjacency; static + dynamic (SDGL). | Discovery; validated against the agronomic graph. |

> **Multiplex, don't pick one.** The stronger contribution is not "which single adjacency wins" but a **multi-relational graph** (correlation layer + directed lead–lag layer + agronomic layer + learned layer) processed by a relational/multi-graph GNN, with an ablation that *removes one relation at a time* to attribute value to each. That is a cleaner scientific claim than a leaderboard of single graphs.

# 4. The Decisive Experiment: Isolating What the Graph Adds

This is the section that determines whether the paper is believed. The claim "the graph helps" is only credible if you **hold model capacity fixed and vary only the edges.** Build a ladder of controls around your best ST-GNN architecture:

1. **Local univariate** (per-crop ARIMA/ETS/LSTM) — ignores all cross-series info.
2. **Global no-graph** — same deep backbone trained across all crops with NO message passing (capacity-matched). Isolates the value of cross-series pooling alone.
3. **Identity-graph GNN** — the GNN with A = I (no neighbors). Sanity floor.
4. **Random / permuted-graph GNN** — same architecture, edges shuffled (degree-preserving). Run many permutations to build a null distribution of accuracy.
5. **Structured-graph GNN** — your real adjacencies (partial-corr, lead–lag, agronomic, learned).

> **The headline result is a single comparison:** does the structured graph beat the **permutation null** by a statistically significant margin (DM test vs the null distribution)? If yes, the topology — not the capacity — carries the signal, and that is a clean, publishable, defensible claim. If the gain is small or insignificant, **that is also publishable** as an honest "when does graph structure actually help search-interest forecasting?" result — top venues value rigorous nuance over hype.

# 5. Evaluation That Survives Review

## 5.1 Metrics

| Metric | Use |
| --- | --- |
| MASE (primary) | Scaled vs seasonal-naïve. The IJF-standard; <1 means you beat the naïve. Replaces MAPE. |
| RMSSE | Scaled squared-error analogue (M5 standard); robust to the 0–100 bound. |
| MAE / RMSE | Report for comparability, but never as the sole evidence. |
| sMAPE | Report only as secondary; note instability near low-interest weeks. |
| Pearson/Spearman of forecast vs actual | Directional/co-movement quality. |
| Pinball loss + interval coverage | If probabilistic (strongly recommended; see §6). |

### 5.2 Protocol and significance

- **Rolling-origin / time-series cross-validation** with multiple origins; report mean ± std across folds and horizons h ∈ {1, 4, 8, 13}.
- **Graphs, scalers, decompositions, and intent-query selection all fit on the training portion of each fold only.** No exceptions.
- **Diebold–Mariano** tests for each model pair; **Friedman + Nemenyi / MCB** across the crop panel for an overall ranking with a critical-difference diagram (the ISF-expected figure).
- **Multiple seeds** for every deep model; report variability, not a single lucky run.
- **Pre-register the crop/keyword list and metrics** before touching test data — directly answers the keyword-selection-bias warning in your reference [23].

## 6. The Robust Pipeline (Reworked)

### 6.1 Establish the target's noise floor first

Before any modeling, quantify how noisy the dependent variable is. Trends resamples on each pull, so y itself varies. Pull each series **N times on different days**, and compute the cross-pull variance. This gives an **irreducible error floor**: no honest model should claim error below it. Reporting skill *relative to the noise floor* is a methodological contribution in its own right and pre-empts the "you're fitting noise" critique.

```
# 6.1 Noise-floor estimation (do this BEFORE modeling)
samples = [fetch_trends(crops, geo='US-FL', freq='weekly') for _ in range(N)]
y_bar   = mean(samples)                  # denoised target
noise   = mean_over_t( var_across_pulls(samples) )   # per-crop noise floor
report(noise)        # any model error below this is overfitting sampling jitter
```

### 6.2 Calibration and stitching

Use **Google Trends Anchor Bank–style calibration** (your reference [20]) to place all crops on a common scale, and joint multi-term queries within anchor groups. Stitch weekly windows onto a monthly backbone (the weekly/monthly switch at ~5 years). Fit every preprocessing step (smoothing, detrending, scaling) on training data only.

### 6.3 Graph construction (per fold, train-only)

```
# 6.3 Validated, deseasonalized, multi-relational graph
R = stl_remainder(panel_train)              # remove trend + annual season
A_part = graphical_lasso(R).precision        # PRIMARY: direct dependence
A_part = statistically_validate(A_part,
             null='stationary_bootstrap', fdr=0.05)   # keep significant edges
A_lag, lag = directed_max_xcorr(R, max_lag=13)        # lead-lag (directed)
A_te       = transfer_entropy(R)                      # nonlinear directed
A_agro     = expert_graph(UF_IFAS, USDA_region, family)
A_learned  = None   # learned end-to-end inside MTGNN/AGCRN/GWN
multiplex  = stack([A_part, A_lag, A_te, A_agro])     # relational GNN input
```

### 6.4 Models and training

Backbone progression: **MTGNN / AGCRN** (learned graph) and **STGCN / DCRNN** (fixed/directed graph), plus **Graph WaveNet** for the learned-vs-fixed story and a **relational/multiplex GNN** for the multi-edge claim. Strong non-graph baselines reviewers now expect: seasonal-naïve, SARIMA/ETS/Theta, VAR, and modern global deep models — **N-HiTS, PatchTST, TFT, DeepAR/TiDE** — via NeuralForecast/Darts. Prefer **probabilistic forecasts** (quantile or DeepAR-style) so you can report pinball loss and coverage; this is well received at ISF and adds a dimension competitors usually skip.

### 6.5 Leakage checklist (expanded from your version)

- Graph, decomposition, scalers, PCA, intent-query selection: train-fold only.
- Causal feature engineering only — rolling stats use past windows; no centered windows.
- Crop/keyword list and metrics pre-registered before test access.
- Same splits, seeds, input window, and horizons across ALL models.
- Archive exact Trends pulls (date, geo, term, window) — the target is not perfectly reproducible, so the raw pulls ARE the dataset.

## 7. Interpretability as Evidence, Not Decoration

Your review already wants "learned edges that make agricultural sense." Make this a **falsifiable test**, not a qualitative observation:

- **Community detection** (Louvain/Leiden) on the learned/validated graph; test alignment with UF-IFAS season classes via adjusted Rand index / modularity vs a label-shuffled null.
- **Learned-vs-agronomic overlap**: quantify edge agreement between Graph WaveNet's adaptive adjacency and the expert agronomic graph (precision/recall of edges).
- **Lead–lag validation**: confirm directed edges (e.g., strawberry→blueberry) with out-of-sample Granger/TE and an ablation that removes the edge and measures the accuracy drop on the downstream crop.

## 8. Reframing the Contribution (Highest-Leverage Change)

> **Forecasting Google Trends better is not, by itself, a top-journal contribution. Forecasting** *demand attention that demonstrably leads real agricultural market outcomes* **is.** The single biggest upgrade is to anchor the search signal to something with stakes.

Three framings, in increasing order of impact. Aim for the third.

| Framing | Claim and what it needs |
|---|---|
| A. Methodological | "When and which graph structure improves multivariate search-interest forecasting." Needs the §4 permutation-null rigor. Publishable on method alone. |
| B. Applied/economic | "Search-interest networks as leading indicators for Florida specialty crops." Needs linkage to USDA-NASS/AMS shipments & prices (Granger/TE from Trends → shipments). |
| C. Both (recommended) | Use the rigorous graph methodology to build VALIDATED leading indicators with demonstrated external validity: graph-based Trends forecasts that nowcast/lead USDA shipment & price movements better than non-graph signals. This is a genuine, defensible, IJF-grade contribution. |

**Concrete external-validity test:** regress (or Granger/TE test) USDA weekly shipment or price series on graph-derived vs non-graph Trends forecasts. If the graph-informed search signal leads real market outcomes — and the permutation null does not — you have shown the network captures economically meaningful demand structure, not just autocorrelation. That sentence is the abstract of a strong paper.

## 9. Summary: Keep / Change / Add

| Verdict | Items |
|---|---|
| KEEP | One-crop-per-node backbone; Design-A start; the edge-type taxonomy; train-only graph construction; the leakage checklist; per-crop reporting; MTGNN/AGCRN/GWN model choices; pre-defining keywords. |
| CHANGE | Raw correlation → deseasonalized PARTIAL correlation + edge validation; MAPE/sMAPE → MASE/RMSSE; 12-wk window → include annual lag/seasonal features; 'GNN vs LSTM' → capacity-matched global baseline + permutation null; single graph → multiplex/relational; point forecasts → probabilistic. |
| ADD | Noise-floor estimation; search-intent node decomposition; transfer entropy; statistical edge validation (FDR); Diebold–Mariano + Friedman/MCB; external validity vs USDA shipments/prices; community-vs-season falsifiable test; multi-seed reporting; data/code archive. |

## 10. Revised Implementation Order

6. **Lock the target.** Pre-register the crop(-intent) keyword list; multi-pull each series; estimate and report the noise floor; calibrate/stitch to a clean weekly panel.
7. **Acquire external-validity data.** Pull USDA-NASS/AMS weekly shipments & prices and FL weather/Wikipedia in parallel — this is what makes the paper matter.
8. **Build validated multi-relational graphs** (partial-corr + lead–lag + TE + agronomic) on training folds; visualize and run the community-vs-season test.

9. **Stand up the full baseline ladder** including the capacity-matched global no-graph model and the permutation-null harness.
10. **Run the decisive experiment (§4)** with DM tests and a critical-difference diagram; report skill relative to the noise floor.
11. **Demonstrate external validity** (Trends-graph forecasts → USDA shipments/prices).
12. **Write for IJF/ISF**: lead with the methodological + economic claim, full reproducibility appendix, honest small-data and noisy-target discussion.

**Stack:** Python · pandas/statsmodels/pmdarima · NeuralForecast & Darts (global deep + probabilistic baselines) · PyTorch Geometric / PyG-Temporal (ST-GNNs) · scikit-learn GraphicalLasso · dtaidistance/tslearn (DTW) · PyIF/IDTxl (transfer entropy) · NetworkX + Leiden (graph + communities) · Google Trends Anchor Bank (calibration) · Wikimedia & USDA QuickStats / AMS APIs.